

Assignment 05

To develop and perform unit testing of smart contracts using the Truffle framework

Objective

Implement a simple Solidity Smart Contract and perform UNIT Testing using Truffle to verify contract functionality.

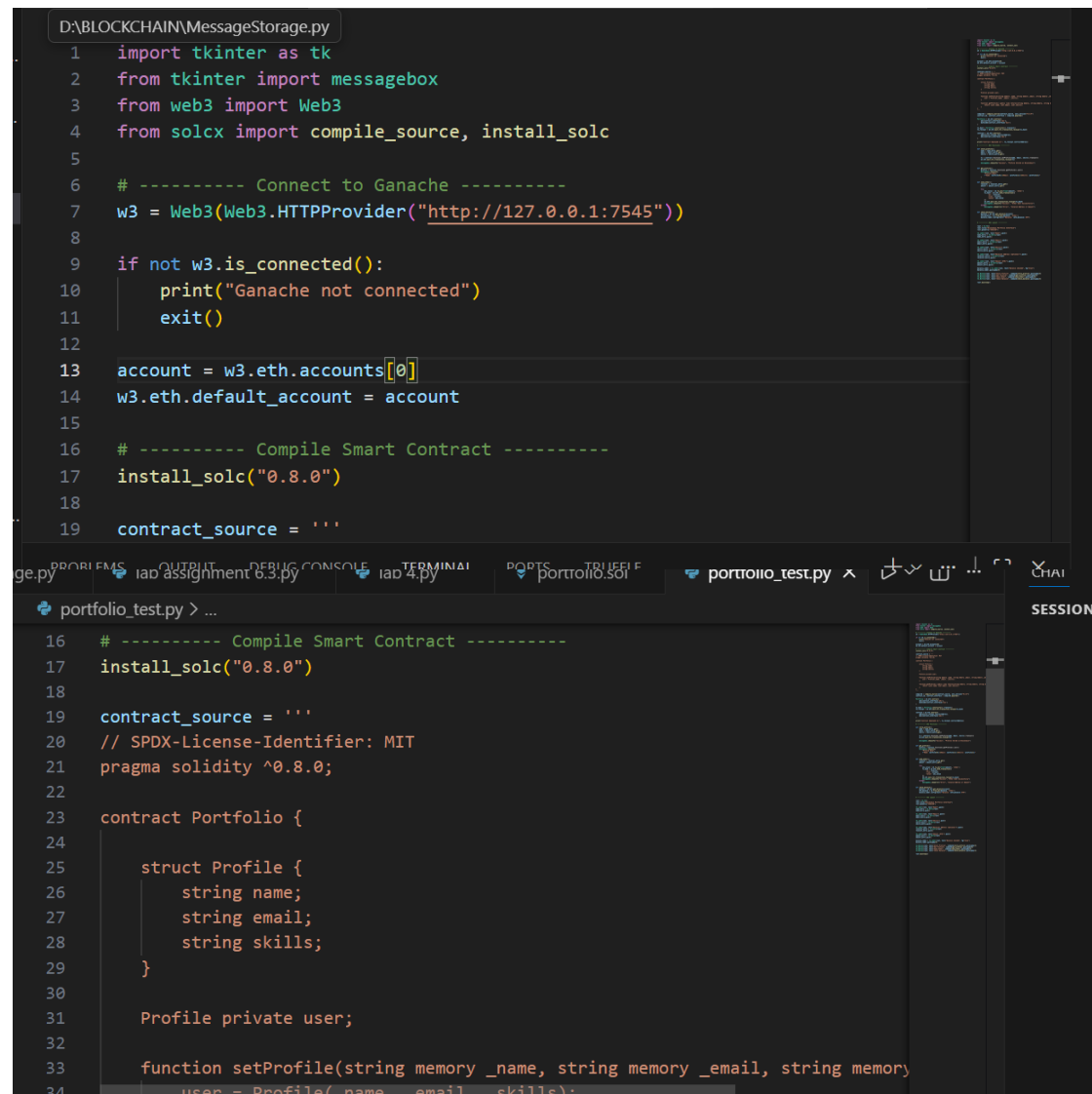
Problem Statement (Equivalent to Given Cipher Practical)

Develop a basic Personal Portfolio Smart Contract that stores and retrieves user profile data, and validate its behavior using Truffle unit tests.

Requirements

Development Environment

- Install Node.js
- Install Truffle Framework
- Install Ganache (Local Blockchain)
- Use VS Code with:
 - o Solidity Extension
 - o JavaScript Extension



```
D:\BLOCKCHAIN\MessageStorage.py
1 import tkinter as tk
2 from tkinter import messagebox
3 from web3 import Web3
4 from solcx import compile_source, install_solc
5
6 # ----- Connect to Ganache -----
7 w3 = Web3(Web3.HTTPProvider("http://127.0.0.1:7545"))
8
9 if not w3.is_connected():
10     print("Ganache not connected")
11     exit()
12
13 account = w3.eth.accounts[0]
14 w3.eth.default_account = account
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16 # ----- Compile Smart Contract -----
17 install_solc("0.8.0")
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19 contract_source = ''
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```

```
35     }
36
37     function getProfile() public view returns(string memory, string memory, string memory) {
38         return (user.name, user.email, user.skills);
39     }
40 }
41 '''
42
43 compiled = compile_source(contract_source, solc_version="0.8.0")
44 contract_id, contract_interface = compiled.popitem()
45
46 Portfolio = w3.eth.contract(
47     abi=contract_interface['abi'],
48     bytecode=contract_interface['bin']
49 )
50
51 tx_hash = Portfolio.constructor().transact()
52 tx_receipt = w3.eth.wait_for_transaction_receipt(tx_hash)
53
54 contract = w3.eth.contract(
55     address=tx_receipt.contractAddress,
56     abi=contract_interface['abi']
57 )
58
59 print("Contract deployed at:", tx_receipt.contractAddress)
60
61 # ----- GUI Functions -----
62
63 def store_profile():
64     name = name_entry.get()
65     email = email_entry.get()
66     skills = skills_entry.get()
67
68     tx = contract.functions.setProfile(name, email, skills).transact()
69     w3.eth.wait_for_transaction_receipt(tx)
70
71     messagebox.showinfo("Success", "Profile Stored on Blockchain")
72
73
74 def get_profile():
75     profile = contract.functions.getProfile().call()
76     messagebox.showinfo(
77         "Profile Data",
78         f"Name: {profile[0]}\nEmail: {profile[1]}\nSkills: {profile[2]}"
79     )
80
81
```

portfolio_test.p

Describe what to build

🔍 ⌵ ⌵ ⌵ Auto ⌵

```
portfolio_test.py > ...
82 def send_ether():
83     receiver = receiver_entry.get()
84     amount = amount_entry.get()
85
86     try:
87         wei_value = w3.to_wei(float(amount), 'ether')
88         tx_hash = w3.eth.send_transaction({
89             'to': receiver,
90             'from': account,
91             'value': wei_value
92         })
93         w3.eth.wait_for_transaction_receipt(tx_hash)
94         messagebox.showinfo("Success", "Ether Sent Successfully")
95     except:
96         messagebox.showerror("Error", "Invalid Address or Amount")
97
98
99 def check_balance():
100     balance = w3.eth.get_balance(account)
101     eth_balance = w3.from_wei(balance, 'ether')
102     balance_label.config(text=f"Balance: {eth_balance} ETH")
103
104 # ----- GUI Layout -----
105
106 root = tk.Tk()
107 root.title("Blockchain Portfolio Interface")
108 root.geometry("350x420")
109
```

```
portfolio_test.py > ...
113 name_entry.pack()
114
115 tk.Label(root, text="Email").pack()
116 email_entry = tk.Entry(root)
117 email_entry.pack()
118
119 tk.Label(root, text="Skills").pack()
120 skills_entry = tk.Entry(root)
121 skills_entry.pack()
122
123 tk.Label(root, text="Receiver Address (optional)").pack()
124 receiver_entry = tk.Entry(root)
125 receiver_entry.pack()
126
127 tk.Label(root, text="Amount (ETH)").pack()
128 amount_entry = tk.Entry(root)
129 amount_entry.pack()
130
131 balance_label = tk.Label(root, text="Balance checked", fg="blue")
132 balance_label.pack(pady=5)
133
134 tk.Button(root, text="Store Profile", command=store_profile).pack(pady=5)
135 tk.Button(root, text="Get Profile", command=get_profile).pack(pady=5)
136 tk.Button(root, text="Send Ether", command=send_ether).pack(pady=5)
137 tk.Button(root, text="Check Balance", command=check_balance).pack(pady=5)
138
139 root.mainloop()
140
```

